How BARCS Finds Important Genes

Chapter 2: The calculations

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Companion slides to the student guide · BARCS 0.2.2 · Grades 7 to 9

The plan for Chapter 2

6. How BARCS finds the best β values
7. How BARCS measures the biological noise
8. Is the effect real?
9. Borrow information from other guides
10. Check the results with control guides
11. From guides to genes
12. Many genes and false discoveries

Remember

Each section has the same title as a step in Chapter 2 of the textbook. The numbers are real BARCS output for our example (0.2.1; unchanged in 0.2.2).

What you need from Chapter 1

Our example and the key ideas

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

- Proportions: 191 (control) and 111 (drug) per million. Fold change 0.58.
- Log-odds: -8.56 (control), -9.10 (drug).
- Variance: $N \times \mu \times (1-\mu) \times \big(1 + (N-1) \times \rho\big)$; $\rho =$ correlation of two reads from one library.
- Independent values: variances add. $\times c$: variance $\times\, c \times c$.
- Design matrix X rows: $(1,0)$, $(1,0)$, $(1,1)$, $(1,1)$.
- Prediction $\eta =$ row of X times β; $\mu = e^{\eta}/(1+e^{\eta})$.
- β_0: control log-odds. β_1: effect of the drug.

Step 6: How BARCS finds the best β values

Guess, check, improve

BARCS uses a loop: iteratively reweighted least squares (IRLS).

1. **Guess.** Start with the simple log-odds of each group.
2. **Predict.** Use X to predict η and μ for each library.
3. **Measure the noise.** Find ρ (Step 7).
4. **Weigh.** Less noise gives a larger weight.
5. **Correct.** Calculate a working value for each library.
6. **Improve.** Weighted averages give the new β values.
7. **Repeat** until the β values stop changing.

The working value: a straight-line step

$\ln(0)$ does not exist, so BARCS uses a straight line for the log-odds.

Part	At $p = 0.2$	At $p = 0.201$	Change $\div$ 0.001
$\ln(p)$	-1.60944	-1.60445	$4.99 \approx 1 \div 0.2 = 5$
$-\ln(1-p)$	$+0.22314$	$+0.22439$	$1.25 = 1 \div 0.8$
log-odds	-1.38629	-1.38006	$6.24 \approx 6.25$

$$\text{slope} = \frac{1}{p} + \frac{1}{1-p} = \frac{(1-p)+p}{p \times (1-p)} = \frac{1}{p \times (1-p)}$$

$$v_i = \eta_i + (K_i/N_i - \mu_i) \times \frac{1}{\mu_i \times (1-\mu_i)}$$

Start at the prediction η_i; step along the line by the distance $K_i/N_i - \mu_i$.

Why weights? Weight $= 1 \div$ variance

Two measurements of the same thing: variances 1 and 4.

Weights	Variance of the average	
equal (1 and 1)	$(1 + 4) \div (2 \times 2) = 1.25$	
$1 \div 1 = 1$ and $1 \div 4 = 0.25$	$1 \div (1 + 0.25) = 0.8$	smaller

Remember

Weight $= 1 \div$ variance gives the average with the smallest variance. So BARCS uses $w_i = 1 \div$ (variance of v_i).

The weight from the variance of the working value

Quantity	Formula	Library 1, iteration 1
1. variance of K_i	$N_i \times \mu_i \times (1-\mu_i) \times \big(1+(N_i-1)\times\rho\big)$	$190.87 \times 1.3841 = 264.19$
2. variance of K_i/N_i	line 1 $\div(N_i \times N_i)$	2.642×10^{-10}
3. variance of v_i	line 2 $\div\big(\mu_i \times (1-\mu_i)\big)^2$	0.007251
4. weight w_i	1$\div$ line 3	137.90

$$w_i = \frac{N_i \times N_i \times \big(\mu_i \times (1-\mu_i)\big)^2}{N_i \times \mu_i \times (1-\mu_i) \times \big(1+(N_i-1)\times\rho\big)} = \frac{N_i \times \mu_i \times (1-\mu_i)}{1+(N_i-1)\times\rho}$$

Line 1: Chapter 1, Step 4. Line 2: divide by $N \Rightarrow$ variance $\div N \times N$. Line 3: the step multiplies by the slope $\Rightarrow$ variance $\times$ slope^2.

New β values, and why a loop

For our design matrix, weighted averages of the working values:

$$\beta_0 = \frac{w_1 \times v_1 + w_2 \times v_2}{w_1 + w_2}, \qquad \beta_0 + \beta_1 = \frac{w_3 \times v_3 + w_4 \times v_4}{w_3 + w_4}$$

Remember

w and v use μ; μ comes from β; β comes from w and v. So BARCS guesses, then repeats until nothing changes.

Iteration 1: guess, predict, measure the noise

Worked example 10a: iteration 1

Guess. Control log-odds -8.563522, drug log-odds -9.104869.
So $\beta_0 = -8.563522$ and $\beta_1 = -9.104869 - (-8.563522) = -0.541347$.

Predict. Control $\mu = 0.000190909$. Drug $\mu = 0.000111111$.

Measure the noise. Step 7 gives $\rho = 0.0000003841$.

Iteration 1: weigh and correct library 1

Worked example 10b: library 1

Weigh.

Top: $1{,}000{,}000 \times 0.000190909 \times 0.999809 = 190.87$

Bottom: $1 + 999{,}999 \times 0.0000003841 = 1.3841$

Weight: $w_1 = 190.87 \div 1.3841 = 137.90$

Correct.

$K_1/N_1 - \mu_1 = 0.000180 - 0.000190909 = -0.0000109$

$-0.0000109 \div 0.000190873 = -0.0572$

Working value: $v_1 = -8.5635 + (-0.0572) = -8.6207$

Iteration 1: improve

Lib.	w_i	v_i	$w_i \times v_i$
1	137.905	−8.62068	−1188.84
2	156.785	−8.51589	−1335.16
3	80.269	−9.02486	−724.42
4	67.988	−9.20488	−625.82

Worked example 10c: new values

Control:

$(-1188.84 - 1335.16) \div 294.69 = -8.564929$

Drug:

$(-724.42 - 625.82) \div 148.257 = -9.107414$

$\beta_1 = -9.107414 - (-8.564929) = -0.542486$

The next iterations

After iteration	β_0	β_1
0 (the guess)	-8.563522	-0.541347
1	-8.564929	-0.542486
2	-8.564934	-0.542492
3	-8.564934	-0.542492

After iteration 3 nothing changes. Final: $\beta_0 = -8.565$, $\beta_1 = -0.542$.

Remember

The weights move β_1 from -0.5413 to -0.5425. Library 2 has the most reads and the least relative noise, so it counts the most.

Step 7: How BARCS measures the biological noise

Pearson residuals and the rule for ρ

A Pearson residual is the distance from the prediction in standard deviations:

$$r_i = \frac{K_i - N_i \times \mu_i}{\sqrt{N_i \times \mu_i \times (1 - \mu_i) \times (1 + (N_i - 1) \times \rho)}}$$

Remember

BARCS chooses ρ so that the squared Pearson residuals add up to the degrees of freedom:

$$r_1^2 + r_2^2 + r_3^2 + r_4^2 = \mathrm{df}$$

Why the sum should be the degrees of freedom

- A Pearson residual is a distance in SDs. If the model is right, its square is 1 on average (variance $\div$ variance).
- True μ: four libraries $\Rightarrow$ the sum is about 4.
- BARCS uses the fitted μ. Each fitted β pulls the prediction toward the data and takes away 1 on average: $4 - 2 = 2 = \text{df}$.

Worked example computer check

20,000 artificial guides like ours, with the true ρ:
true μ: average sum 4.00; fitted μ: average sum 2.00.

So BARCS chooses ρ to make the sum exactly df (a *method of moments*).

Start with $\rho = 0$

Worked example 11

Library	$(K - N \times \mu)^2$	$N \times \mu \times (1 - \mu)$	Squared residual
1	$(-10.64)^2 = 113.21$	190.60	0.594
2	$(+11.23)^2 = 126.16$	228.72	0.552
3	$(+9.17)^2 = 84.14$	110.82	0.759
4	$(-8.66)^2 = 75.03$	88.65	0.846
		Sum	2.751

The sum (2.751) is bigger than df $= 2$. So there is more noise than counting noise: $\rho > 0$.

Guess and check for ρ

ρ	Sum
0	2.751
0.0000001	2.506
0.0000002	2.302
0.0000003	2.129
0.000000386	2.000
0.0000005	1.851

- A larger ρ makes the sum smaller.
- BARCS stops when the sum equals 2: $\rho = 0.000000386$.
- If the sum is 2 or less at $\rho = 0$, BARCS uses $\rho = 0$.

Variance inflation

$2.751 \div 2 = 1.376$: the guide has 1.38 times more noise than counting noise alone (used in Step 9).

Step 8: Is the effect real?

The standard error

The standard error (SE) tells how much β_1 can change because of noise. For our design matrix, it comes from the weights:

$$W_{\text{control}} = w_1 + w_2, \quad W_{\text{drug}} = w_3 + w_4, \qquad \text{SE}(\beta_1) = \sqrt{\frac{1}{W_{\text{control}}} + \frac{1}{W_{\text{drug}}}}$$

Worked example 13

Final weights: 137.524, 156.322, 79.955, 67.737.

$W_{\text{control}} = 293.846, \quad W_{\text{drug}} = 147.692$

$1/293.846 + 1/147.692 = 0.003403 + 0.006771 = 0.010174$

$\text{SE} = \sqrt{0.010174} = 0.101$

Where the SE formula comes from

Step	Why
1. variance of $v_i = 1 \div w_i$	definition of the weight
2. $\beta_0 = \frac{w_1}{W_c} \times v_1 + \frac{w_2}{W_c} \times v_2$	Step 6 ($W_c = w_1 + w_2$)
3. var. of $\beta_0 = \left(\frac{w_1}{W_c}\right)^2 \times \frac{1}{w_1} + \left(\frac{w_2}{W_c}\right)^2 \times \frac{1}{w_2}$	$\times c \Rightarrow$ variance $\times\, c \times c$; add
4. $= \frac{w_1+w_2}{W_c \times W_c} = \frac{1}{W_c}$	one w cancels in each part
5. var. of $(\beta_0 + \beta_1) = 1 \div W_{\text{drug}}$	same for the drug libraries
6. var. of $\beta_1 = 1 \div W_c + 1 \div W_{\text{drug}}$	difference; $(-1) \times (-1) = +1$, so variances add
7. SE $= \sqrt{\text{line 6}} = \sqrt{0.003403 + 0.006771} = 0.101$	SE = SD of the estimate

The t value and the degrees of freedom

$$t = \frac{\beta_1}{\text{SE}}, \qquad \text{df} = (\text{rows of } X) - (\text{columns of } X)$$

- A t value far from 0: the effect is large compared with the noise.
- Fewer degrees of freedom: BARCS needs a bigger t.

Degrees of freedom	1	2	14	very many
$\lvert t \rvert$ needed for $p < 0.05$	12.71	4.30	2.14	1.96

The p-value

- The p-value: if the true effect is 0, how often does noise alone give a t this far from 0?
- Small p-value: noise alone seldom gives such a result. The usual limit is 0.05.

Worked example 14: testing the effect

1. $t = -0.542 \div 0.101 = -5.38$
2. X has 4 rows and 2 columns: $\text{df} = 2$
3. $\text{df} = 2$ needs $|t| > 4.30$. Our $|t| = 5.38$. So $p < 0.05$.
4. Exact p-value: $p = 0.033$

Degrees of freedom come from libraries

Remember

The degrees of freedom come from the number of libraries, not from the number of reads.

A library with 100 million reads still counts as one row of X.

Step 9: Borrow information from other guides

Moderation: a weighted vote

- 2 degrees of freedom give an unreliable noise estimate.
- A screen has thousands of guides with similar noise.
- BARCS averages the guide's own inflation with the typical inflation.

$$\text{moderated} = \frac{\text{df} \times \text{own} + d_0 \times \text{typical}}{\text{df} + d_0}$$

What the symbols mean

d_0: the prior degrees of freedom, the votes of the typical value. BARCS calculates it from all the guides.

Worked example: moderation with simple numbers

Worked example 15

A guide has df $= 3$ and own inflation 4.0. Typical inflation 1.5 with $d_0 = 12$.

$$\frac{3 \times 4.0 + 12 \times 1.5}{3 + 12} = \frac{12 + 18}{15} = \frac{30}{15} = 2.0$$

The value moves from 4.0 toward 1.5 and stops at 2.0.

What moderation does to our guide

BARCS uses the moderated value to:

1. multiply the SE by $\sqrt{\text{moderated} \div \text{own}}$ (variance $\propto$ inflation; SE $= \sqrt{\text{variance}}$; own below 1 counts as 1)
2. recalculate t
3. use df $+ d_0$ degrees of freedom

BARCS moderates when it has 50 or more usable guides.

Worked example 16

Own inflation 1.376, df $= 2$. Typical 1.5, $d_0 = 12$.
Moderated: $(2 \times 1.376 + 12 \times 1.5) \div 14 = 1.482$
SE factor: $\sqrt{1.482 \div 1.376} = 1.038$
New SE: $0.101 \times 1.038 = 0.105$
New t: $-0.542 \div 0.105 = -5.18$, df $= 14$
New p-value: 0.00014

Step 10: Check the results with control guides

Non-targeting control guides

- Control guides switch off no gene. Their true effect is 0.
- About 5% of them should get $p < 0.05$ by chance.
- More than that: the noise model is too small. BARCS makes the tests wider.

Worked example 17: the scale factor

100 control guides: we expect 5 with $p < 0.05$, but 15 have it.
95 of the 100 control guides have $|t|$ below 2.47. The table says 1.96.
Scale factor: $2.47 \div 1.96 = 1.26$. BARCS divides every t by 1.26: $t = 3.5$ becomes $3.5 \div 1.26 = 2.78$.

The scale factor is never less than 1. Calibration only makes results more careful.

Step 11: From guides to genes

The Stouffer method

1. Change each guide's p-value into a z score. The sign follows the effect.
2. Add the z scores.
3. Divide by the square root of the number of guides, m.

$$Z_{\text{gene}} = \frac{z_1 + z_2 + \cdots + z_m}{\sqrt{m}}$$

Why $\sqrt{m}$: each z has variance 1, so the sum has variance m; dividing by $\sqrt{m}$ divides the variance by $m \Rightarrow$ variance 1 again.

Two-sided p-value	0.317	0.134	0.046	0.012	0.0027
z score	1.0	1.5	2.0	2.5	3.0

Worked examples: combining guides

Worked example 18: three guides

p-values 0.046, 0.134, 0.012 give $z = 2.0$, 1.5, 2.5.

$$Z = \frac{2.0 + 1.5 + 2.5}{\sqrt{3}} = \frac{6.0}{1.732} = 3.46$$

$p = 0.00053$. Together, the guides give strong evidence.

Worked example 19: shared noise

Guides can share noise (guide correlation r). Squares and pairs (Chapter 1): m squares $+$ $m \times (m-1)$ pairs $\times r$. Bottom: $\sqrt{m + m \times (m-1) \times r}$.
With $r = 0.1$: $\sqrt{3 + 6 \times 0.1} = \sqrt{3.6} = 1.897$.
$Z = 6.0 \div 1.897 = 3.16$, $p = 0.0016$.

Step 12: Many genes and false discoveries

Step 13: Let the control guides decide (BARCS 0.2.2)

When scaling is not enough: pseudo-genes

Sometimes control guides have more extreme values than any scale factor fixes (e.g. flasks split from one population share large clones). Idea: do not trust the model for "no effect". Ask the control guides.

Pseudo-gene: pick m control guides at random (as many as the gene has), average their signed z scores. Repeat, e.g. 999 times.

$$p = 2 \times \frac{1 + (\text{pseudo-genes at least as far out, same side})}{1 + (\text{number of pseudo-genes})}$$

Example 21: gene with 4 guides, average z $= +2.4$; 4 of 999 pseudo-genes reach $+2.4$: $p = 2 \times (1 + 4) \div (1 + 999) = 0.01$. Then Benjamini–Hochberg.

Shared noise and limits

Widen for shared noise. Random controls do not share noise; a real gene's guides do (r). Squares and pairs:

$$\text{variance of the average of } m = \frac{m + m \times (m-1) \times r}{m \times m} = \frac{1}{m} \times \big(1 + (m-1) \times r\big)$$

So BARCS widens each pseudo-gene by $\sqrt{1 + (m-1) \times r}$ ($m = 4$, $r = 0.2$: $\sqrt{1.6} = 1.26$).

Limits.

- Smallest p-value $= 2 \div (n+1)$: 1,000 controls give 0.002. Over 20,000 one-guide genes, BH needs about 400 genes at that floor to call any at FDR 0.10. "No hits" can mean "too few controls".
- Knockout screens: controls make no cut and can sit apart; `centre = "targets"` takes the centre from the targeting guides.
- Optional; needs at least 100 usable controls.

Why we need a false discovery rate

- 20,000 genes with $p < 0.05$: about 1,000 pass by chance.
- False discovery rate (FDR): the expected fraction of false results in the list of hits.

Idea: with no effect, p-values are spread evenly from 0 to 1. Of 20,000 such genes, about $20{,}000 \times 0.001 = 20$ get $p \leq 0.001$. If 100 genes have $p \leq 0.001$, about $20 \div 100 = 0.2$ of them are false.

Benjamini–Hochberg method:

1. Sort the genes by p-value and give them ranks.
2. Calculate $p \times$ (number of genes) $\div$ rank.
3. From the bottom up, replace a value with the value below it if that value is smaller.

Worked example: FDR for five genes

Worked example 20

Gene	Rank	p	$p \times 5 \div \text{rank}$	FDR
A	1	0.001	$0.001 \times 5 \div 1 = 0.005$	0.005
B	2	0.008	$0.008 \times 5 \div 2 = 0.020$	0.020
C	3	0.039	$0.039 \times 5 \div 3 = 0.065$	0.051 (from D)
D	4	0.041	$0.041 \times 5 \div 4 = 0.051$	0.051
E	5	0.600	$0.600 \times 5 \div 5 = 0.600$	0.600

FDR < 0.05: genes A and B. FDR < 0.10: genes A, B, C and D.

Two special cases

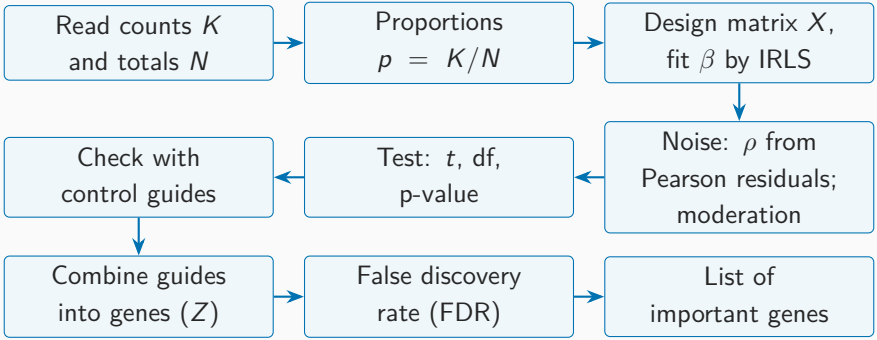
Zero reads and very large changes

A guide with zero reads
0 reads in all control libraries means log-odds $= \ln(0)$, which does not exist. Older BARCS removed the guide. BARCS 0.2.1 first adds less than one read (the Firth correction) and keeps it.

A very large change
A guide that grows hundreds of times can make the usual test fail: the SE grows faster than the effect. The likelihood-ratio test compares the fit with and without the effect. `test = "auto"` uses it only for such guides.

Summary: the full path

From counts to a list of genes



The symbols of Chapter 2

Symbol	Meaning	Value in our example
w	weight of a library in the fit	137.5 (library 1)
working value	prediction plus correction	-8.621 (library 1)
Pearson residual	distance from the prediction in SDs	-0.65 (library 1)
inflation	noise compared with counting noise	1.376
SE	standard error of the effect	0.101
t	effect divided by SE	-5.38
df	rows of X minus columns of X	$4 - 2 = 2$
p-value	chance of this result if the effect is 0	0.033
d_0	prior degrees of freedom (votes)	12
z, Z	z score of a guide, of a gene	3.46
m	number of guides for a gene	3
r	guide correlation (shared noise)	0.1
FDR	expected fraction of false hits	0.005 for gene A